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hw3_me501_Q6.m

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```
close all;
clc;
```

```
disp('Numerically finding equilibrium points for the bendy spring-pendulum.')
```

Numerically finding equilibrium points for the bendy spring-pendulum.

Numerically find the roots of $f(x)=0$

```
% We will use two different guesses:
```

```
q_guess1 = [0.5; 0.5];
q_guess2 = [-0.5; 0.5];
```

```
% find the "closest" equilibrium point to each guess. print them.
```

```
q_eq1 = fsolve(@soln_bendyspringeq_2d, q_guess1)
q_eq2 = fsolve(@soln_bendyspringeq_2d, q_guess2)
```

```
% as position vectors... cartesian
```

```
r_eq1 = [0; q_eq1];
r_eq2 = [0; q_eq2];
```

Equation solved.

fsolve completed because the vector of function values is near zero as measured by the value of the function tolerance, and the problem appears regular as measured by the gradient.

```
q_eq1 =
```

```
    0.3057
    0.3631
```

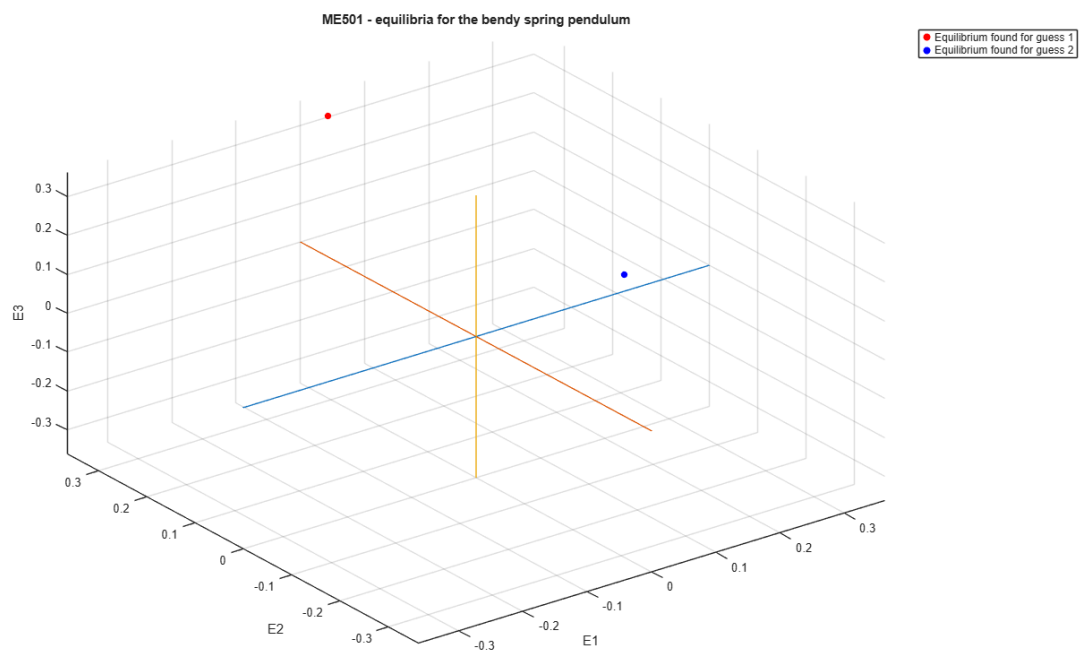
Equation solved.

fsolve completed because the vector of function values is near zero as measured by the value of the function tolerance, and the problem appears regular as measured by the gradient.

```
q_eq2 =  
  
    -0.3057  
     0.3631
```

Plot

```
% set up the figure  
figure; hold on;  
grid on;  
  
title('ME501 - equilibria for the bendy spring pendulum')  
  
xlabel('E1');  
ylabel('E2');  
zlabel('E3');  
% find plot limits based on the particle position vectors we want to  
% plot  
r_all = [r_eq1, r_eq2];  
maxcoord = max([max(r_all), -min(r_all)]);  
limits = [-maxcoord, maxcoord];  
xlim(limits);  
ylim(limits);  
zlim(limits);  
% plot some lines to visualize the E1, E2, E3 axes  
line(2*xlim, [0,0], [0,0]);  
line([0,0], 2*ylim, [0,0]);  
line([0,0], [0,0], 2*zlim);  
% look at the plot from a nice angle  
view(-37, 40);  
  
scatter3(r_eq1(1,:), r_eq1(2,:), r_eq1(3,:), 30, "r", "filled");  
scatter3(r_eq2(1,:), r_eq2(2,:), r_eq2(3,:), 30, "b", "filled");  
legend("", "", "", "Equilibrium found for guess 1", "Equilibrium found for  
guess 2");
```



Published with MATLAB® R2026a